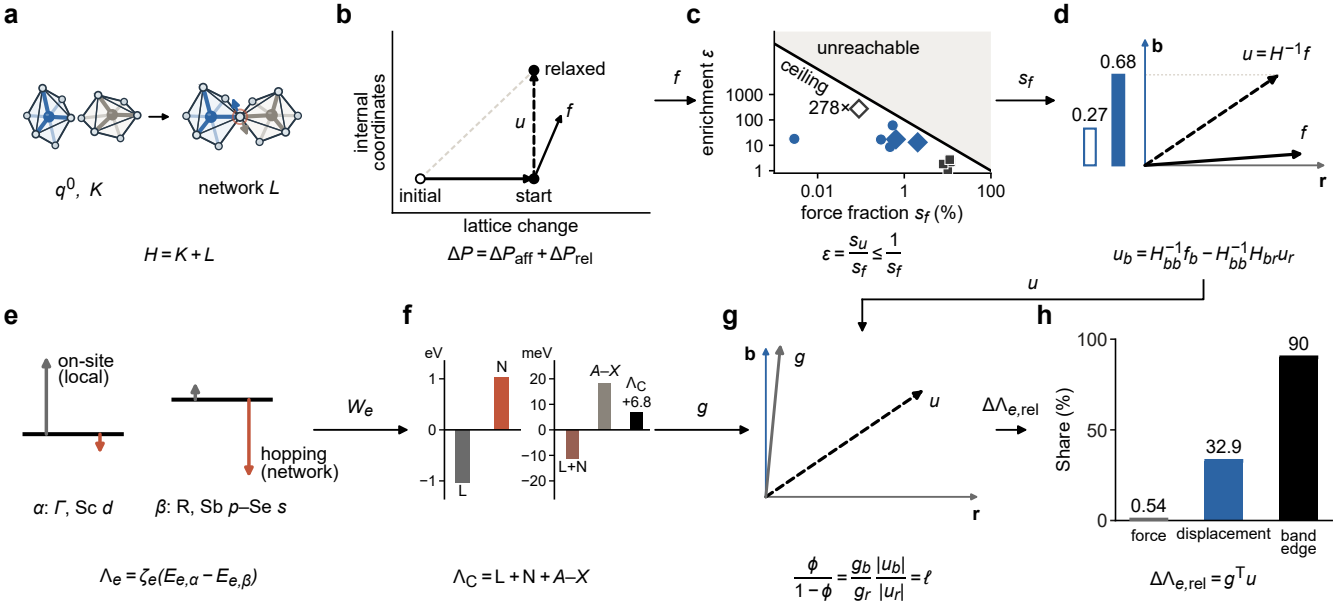

a
q^0, K
network L
$H = K + L$
b
internal coordinates
relaxed
u
f
initial
start
lattice change
$\Delta P = \Delta P_{\text{aff}} + \Delta P_{\text{rel}}$
f
c
enrichment ε
unreachable
ceiling
278×
1000
100
10
1
0.01
1
100
force fraction s_f (%)
$\varepsilon = \frac{s_u}{s_f} \le \frac{1}{s_f}$
s_f
d
b
0.68
0.27
$u = H^{-1} f$
f
r
$u_b = H_{bb}^{-1} f_b - H_{bb}^{-1} H_{br} u_r$
u
e
on-site (local)
hopping (network)
α: Γ, Sc d
β: R, Sb p–Se s
W_e
$\Lambda_e = \zeta_e (E_{e,\alpha} - E_{e,\beta})$
f
eV
meV
N
L
A–X
Λ_C
+6.8
L+N
g
$\Lambda_C = \mathrm{L} + \mathrm{N} + A\text{–}X$
g
b
g
u
r
$\frac{\phi}{1-\phi} = \frac{g_b}{g_r} \frac{|u_b|}{|u_r|} = \ell$
$\Delta\Lambda_{e,\text{rel}}$
h
Share (%)
100
50
0
0.54
32.9
90
force
displacement
band edge
$\Delta\Lambda_{e,\text{rel}} = g^{\mathsf{T}} u$